

VENUGOPAL INDRAJITH HEMASHREE

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EDUCATION

Vellore Institute of Technology, Chennai

Aug 2023 – Present

B.Tech, Computer Science Engineering — Cyber Physical Systems | CGPA: 7.81

Maharishi International Residential School, Kanchipuram

Class 12 (CBSE): 79% | Class 10 (CBSE): 89%

PATENT

IoT-Enabled Smart Walker with Sensor-Fusion Fall Risk Prediction and Adaptive Mobility Assistance

Co-Inventor | Status: **Filing in Progress** (Inventive disclosure filed with IIPRD, Khurana & Khurana) | Faculty Guide: Dr. Subitha D

- Co-invented an Arduino Uno-based smart walker combining dual MPU6050 IMUs, dual HX711 load cells, and dual HC-SR04 ultrasonic sensors for real-time bilateral fall detection, weight-asymmetry monitoring, and obstacle avoidance, running fully standalone with no cloud dependency at an estimated hardware cost under USD 30.
- Designed a sensor-fusion Fall Risk Prediction engine and a Dynamic Stability Reserve metric that combine tilt, weight imbalance, and obstacle proximity to flag deteriorating stability before a fall occurs, alongside near-fall detection and a Gait Symmetry Index for rehabilitation tracking.

PROJECTS

AI-Based Dental Malocclusion Detection and Teeth Alignment Analysis

Jul 2025 – Sep 2025

Python, YOLOv9, Grad-CAM, Gradio, OpenCV, Deep Learning, Jupyter Notebook, GitHub

- Built a YOLOv9-based detection pipeline for automated dental malocclusion and teeth-alignment analysis from X-ray images, achieving **~97% detection accuracy across a dataset of 13,000+ images**.
- Applied Grad-CAM to visualize the regions driving each prediction, supporting clinical interpretability and orthodontic decision support.
- Deployed an interactive Gradio interface for real-time image upload, prediction, and result visualization.

Reinforcement Learning-Driven Autonomous Rover for Hazard Detection and Environmental Exploration

2026

Team Project (3 members) | Python, Reinforcement Learning, Q-Learning, CNN, Simulation

- Built a simulation in which an autonomous rover uses reinforcement learning to navigate a disaster-hit area and map the full space without any pre-given hazard map, discovering hazards through on-the-fly perception instead.
- Implemented tabular Q-learning over a discretized grid environment, paired with a lightweight CNN-based hazard classifier trained on synthetic image data so each grid cell's risk is inferred visually rather than hand-labeled.
- Ran a 4-configuration reward ablation study (baseline / coverage / curiosity / full risk-aware) that confirmed the intended trade-off between area coverage and hazard-safety.
- Produced a step-by-step visual simulation of the rover scanning neighboring cells, updating a live hazard/safe map, and selecting its next move via learned Q-values until the grid is fully mapped.

Intrusion Detection and Threat Modeling

Nov 2025 – Jan 2026

Python, IDS, Threat Modeling, Risk Analysis, Cybersecurity, GitHub

- Designed an intrusion detection system for a simulated online banking platform, using behavioral analysis and risk-based scoring across login, access, and transaction events to flag anomalies.
- Built threat classification logic with automated response triggers, backed by detailed security logging for audit trails and post-incident analysis.

Jarvis — Local Voice-Controlled AI Assistant

Personal Project

Python, SpeechRecognition, Local System Commands

- Built a locally-run Python voice assistant that recognizes spoken commands to open applications and execute system controls (including shutdown/restart), with no cloud dependency.
- Added battery/system-status reporting and stock-price notifications, exploring fully offline voice-driven control end to end.

INTERNSHIP

Research Intern — VIT Chennai, under Prof. Pavithra

May 2025 – Jul 2025 (1 month)

Human Pose Estimation using Heatmaps (MMPose Validation) | MMPose, OpenCV, Kinovea, Matplotlib, Python

- Implemented human joint detection using heatmap-based pose estimation models with MMPose, visualizing and analyzing joint-confidence heatmaps for keypoint localization.
- Validated pose-estimation output against ground truth using Kinovea motion-analysis software to sanity-check keypoint accuracy under faculty supervision.

TECHNICAL SKILLS

Languages: Java, Python

Web Development: HTML, CSS, JavaScript

ML / Computer Vision: YOLOv8/v9, MMPose, OpenCV, Grad-CAM, TensorFlow/PyTorch, Heatmap-based Keypoint Detection

Cybersecurity: Intrusion Detection, Threat Modeling, Risk Analysis

Embedded / IoT: Arduino, Sensor Fusion

Databases & Tools: MySQL, Git, VS Code, Jupyter Notebook, Google Colab, Gradio

Spoken Languages: Tamil (Native), English (Fluent), Hindi (Intermediate), Japanese (Basic)

ACHIEVEMENTS & LEADERSHIP

- **IBM Generative AI Certification** — “Gen AI Using IBM Watsonx,” IBM Career Education Program, via IBM Developer Skills Network (Jun 2025). | [View certificate](#)
- **NUS Introduction to Industry 4.0 Programme, National University of Singapore** (Jun 2025) — completed via the School of Continuing and Lifelong Education (SCALE) & College of Design and Engineering; team member on a final project on digital transformation using the Business Model Canvas. | [View certificate](#)
- **SuperAI Conference 2025, Marina Bay Sands, Singapore** (Jun 2025) — attended Asia's largest AI conference, featuring 7,000+ innovators and 1,000+ AI companies.
- Team Leader, CodeYGen Club — led a student team through coding and design-based projects, coordinating development and presentations.
- Design Lead, Fitness Club — designed promotional creatives, posters, and digital content to support fitness events and member engagement.